

Julian Bellavita

COMPUTER SCIENCE PHD STUDENT AT CORNELL UNIVERSITY · DOE CSGF FELLOW

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Education

Cornell University

PHD, COMPUTER SCIENCE

- Advisor: Dr. Giulia Guidi
- Concentration: Scientific Computing
- GPA: 3.95

Ithaca, NY
08/2023 - present

University of California, Berkeley

BA, COMPUTER SCIENCE

- Major GPA: 3.97

Berkeley, CA
08/2019 - 05/2023

Research and Professional Experience

Cornell University – Graduate Researcher

ADVISOR: DR. GIULIA GUIDI

Large-scale multi-GPU clustering with sparse linear algebra. Parallelizing sparse linear algebra kernels. Mixed precision clustering.

Ithaca, NY
08/2023 - Present

Apple – Intern

VECTOR AND NUMERICS GROUP

Worked on sparse direct and iterative solvers.

Cupertino, CA
05/2026 - 08/2026

Oak Ridge National Laboratory – Graduate Researcher

ADVISOR: DR. RAMAKRISHNAN KANNAN

Mixed precision tensor processing kernels on GPUs.

Oak Ridge, TN
05/2025 - 08/2025

University of Trento – Visiting PhD Student

ADVISOR: DR. FLAVIO VELLA

Kernel k-means on GPUs with sparse linear algebra.

Trento, Italy
01/2024 - 08/2024

University of California, Berkeley – Undergraduate Researcher

ADVISOR: DR. JAMES DEMMEL

Generating BLAS 3 kernels for vector architectures via user-scheduled compilers.

Berkeley, CA
05/2022 - 05/2023

Lawrence Berkeley National Lab – Student Research Assistant

PAGODA PROJECT – CLASS GROUP

Task-based formulation of sparse Cholesky decomposition with GPU acceleration.

Berkeley, CA
08/2022 - 08/2023

Lawrence Berkeley National Lab – Student Research Assistant

SCIENTIFIC DATA MANAGEMENT GROUP

Studying access patterns of high-energy physics datasets on distributed storage systems.

Berkeley, CA
09/2021-05/2023

Publications

Julian Bellavita*, Lorenzo Pichetti*, Thomas Pasquali, Flavio Vella, and Giulia Guidi. 2026. "Communication-Avoiding SpGEMM via Trident Partitioning on Hierarchical GPU Interconnect". In The 40th ACM International Conference on Supercomputing (**ICS 2026**). *Equal contribution

Julian Bellavita, Matthew Rubino, Nakul Iyer, Andrew Chang, Aditya Devarakonda, Flavio Vella, and Giulia Guidi. 2026. "Communication-Avoiding Linear Algebraic Kernel K-Means on GPUs". In The 40th IEEE International Parallel and Distributed Processing Symposium (**IPDPS 2026**).

Julian Bellavita, Thomas Pasquali, Laura Del Rio Martin, Flavio Vella, and Giulia Guidi. 2025. "Popcorn: Accelerating Kernel Kmeans on GPUs through Sparse Linear Algebra". In The 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (**PPoPP 2025**).

Thomas McFarland, **Julian Bellavita**, and Giulia Guidi. 2025. "Parallel GPU-Enabled Algorithms for SpGEMM on Arbitrary Semirings with Hybrid Communication". Short paper. In Proceedings of the 16th ACM/SPEC International Conference on Performance Engineering (**ICPE 2025**).

Adrián Castelló, **Julian Bellavita**, Grace Dinh, Yuka Ikarashi, Hector Martínez. 2024. "Tackling the Matrix Multiplication Micro-Kernel Generation with Exo." In IEEE/ACM International Symposium on Code Generation and Optimization (**CGO 2024**).

Julian Bellavita, Mathias Jacquelin, Esmond G. Ng, Dan Bonachea, Johnny Corbino, and Paul H. Hargrove. 2023. "symPACK: A GPU-Capable Fan-Out Sparse Cholesky Solver." In Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis (**SC-W 2023**).

Julian Bellavita, Caitlin Sim, Kesheng Wu, Alex Sim, Shinjae Yoo, Hiro Ito, Vincent Garonne, Eric Lancon. 2023. "Understanding Data Access Patterns for dCache System." In 26th International Conference on Computing in High Energy & Nuclear Physics (**CHEP 2023**)

Julian Bellavita, Alex Sim, Kesheng Wu, Inder Monga, Chin Guok, Frank Würthwein, and Diego Davila. 2022. "Studying Scientific Data Lifecycle in On-demand Distributed Storage Caches." In Fifth International Workshop on Systems and Network Telemetry and Analytics (**SNTA 2022**).

Awards

FELLOWSHIPS

2024 **DOE Computational Science Graduate Fellowship (DOE CSGF)**, United States Department of Energy

2023 **Cornell Fellowship**, Cornell University

AWARDS

2024 **Best Student Presentation**, The 2nd CINI Summer School on High Performance Computing and Emerging Technologies

2022 **2nd Place**, ACM/IEEE Student Research Competition, Undergraduate Division

GRANTS

2025 **NSF Student Travel Grant**, The 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP 2025)

Presentations

TALKS

"Block Leverage Scores for Cache-Friendly Randomized CP Decomposition", RNLA Workshop at IPAM, 2025

"Algorithms for Computing a Tensor Times Matrix Chain in Mixed Precision", Oak Ridge National Laboratory Discrete Algorithms Group Seminar, 2025

"Multi-GPU Communication Schemes for Large-Scale Supercomputers", Cornell Systems Seminar, 2024

"Accelerating High-Dimensional K-Means Clustering on GPUs with Sparse Matrix Multiplication", University of Trento, 2024

"RDMA-Based Algorithms for Sparse Matrix Multiplication on GPUs", Cornell HPC Group, 2024

"Portable code generation and semi-automatic scheduling for BLIS microkernels with Exo", BLIS Retreat, 2022

POSTERS

"Forward Error Bounds and Efficient Algorithms for Computing a Tensor Times Matrix Chain in Low Precision on GPUs", ACM Student Research Competition at SC, 2025

"Mixed Precision Algorithms for Computing the Tucker Decomposition", CSGF Program Review, 2025

"Efficient Large-Scale Multi-GPU Clustering using Sparse Linear Algebra", IPDPS PhD Forum, 2025

Skills

Languages: C, C++, Python, Julia, Fortran

Frameworks/Libraries: CUDA, MPI, OpenMP, NVSHMEM, UPC++, Metal, Matplotlib, NumPy, Pandas